

Weekly Report: PA-CSI DG LOEO Ablation

April 15, 2026

1 Progress

1.1 Experiment Status

This week focused on the leave-one-environment-out (LOEO) ablation study for the PA-CSI domain generalization pipeline. The current paper-aligned single-seed experiment queue contains five ablations:

1. baseline
2. SupCon only
3. NARC only
4. SupCon + NARC
5. SupCon + NARC + AdaIN

At the time of writing, all five ablations have completed for all three held-out environments ($E1$, $E2$, $E3$), including SupCon + NARC + AdaIN.

1.2 Current Comparable LOEO Ablation Results

Table 1 reports the completed paper-aligned LOEO ablations. These are the current canonical results because they share the same data split protocol, paper-aligned backbone, optimizer schedule, and training duration.

Table 1: Completed paper-aligned LOEO ablations (seed 42).

Experiment	Mean Acc.	Mean Macro-F1	Note
Baseline	0.8787	0.8515	CE only
SupCon only	0.9027	0.8820	Best completed result
NARC only	0.8616	0.8377	Worse than baseline
SupCon + NARC	0.8844	0.8617	Better than NARC only, below SupCon only
SupCon + NARC + AdaIN	0.8899	0.8618	Best combined objective, still below SupCon only

Table 2: Per-environment LOEO accuracy for completed paper-aligned ablations.

Experiment	Target $E1$	Target $E2$	Target $E3$
Baseline	0.9007	0.8547	0.8807
SupCon only	0.9147	0.8953	0.8980
NARC only	0.8967	0.8620	0.8260
SupCon + NARC	0.8813	0.8830	0.8890
SupCon + NARC + AdaIN	0.9103	0.8737	0.8857

1.3 Class-wise Results

The six classes in the local preprocessing pipeline are `bed`, `fall`, `run`, `sitdown`, `standup`, and `walk`. Table 3 reports the mean class recall averaged over the three held-out environments for the completed paper-aligned ablations.

Table 3: Mean class recall across LOEO targets $E1$, $E2$, and $E3$ for completed paper-aligned ablations.

Experiment	bed	fall	run	sitdown	standup	walk
Baseline	0.9986	0.5450	0.8292	0.9475	0.7742	0.9967
SupCon only	0.9869	0.5908	0.9158	0.9550	0.8492	0.9967
NARC only	0.9672	0.5367	0.7908	0.9192	0.8142	0.9983
SupCon + NARC	0.9736	0.5708	0.8958	0.9258	0.8225	0.9950
SupCon + NARC + AdaIN	0.9850	0.4592	0.9225	0.9800	0.8608	0.9933

1.4 AdaIN Update

The `SupCon + NARC + AdaIN` run is now complete. Its final mean accuracy is 0.8899 and its mean macro-F1 is 0.8618. This is slightly better than `SupCon + NARC`, but it still does not exceed `SupCon` only.

Table 4: Final LOEO results for `SupCon + NARC + AdaIN`.

Target Environment	Accuracy	Macro-F1
$E1$	0.9103	0.8817
$E2$	0.8737	0.8522
$E3$	0.8857	0.8515
Mean ($E1$ – $E3$)	0.8899	0.8618

1.5 All Existing LOEO Outputs in the Repository

In addition to the current canonical paper-aligned runs, the repository also contains earlier exploratory LOEO outputs. These are useful as progress milestones, but they should not be mixed with the paper-aligned table because the training setup changed across iterations.

Table 5: Earlier exploratory LOEO runs already present in the repository.

Experiment Output	Mean Acc.	Mean Macro-F1	Scope
<code>diag_single_seed / baseline_pa_csi_dg</code>	0.8037	0.7760	Old baseline, full $E1/E2/E3$
<code>single_seed_fixed / pa_csi_dg_supcon_narc</code>	0.8372	0.8109	Earlier SupCon+NARC, full $E1/E2/E3$
<code>sanity / pa_csi_dg_supcon_narc</code>	0.6940	0.6498	Sanity-only run, target $E1$ only

2 Analysis

2.1 Overall

The completed canonical ablations show a stable ranking:

SupCon only > SupCon + NARC + AdaIN > SupCon + NARC > Baseline > NARC only

for both mean accuracy and mean macro-F1. The main reason is that SupCon promotes class-level clustering across environments, while the current NARC formulation mainly promotes sample-level consistency across antenna views.

2.2 Class-wise Takeaways

The class table adds three useful observations:

- The best class-wise performers are now split across methods: `bed` is highest under Baseline (0.9986), `fall` under SupCon only (0.5908), `run/sitdown/standup` under SupCon + NARC + AdaIN (0.9225/0.9800/0.8608), and `walk` under NARC only (0.9983).
- `fall` remains the hardest class across all completed ablations, and AdaIN hurts it most, reducing mean recall to 0.4592.
- AdaIN improves several medium-difficulty classes, but those gains are not large enough to overturn the overall ranking because SupCon only remains more balanced across classes.

2.3 Why NARC Is Weaker Here

NARC only does not help the classes that matter most for the ranking. It slightly improves saturated classes such as `walk`, but it reduces average recall on `run`, `fall`, and `sitdown` relative to SupCon only. The clearest example is target $E3$, where `run` recall drops from 0.9425 under SupCon only to 0.6300 under NARC only. This is consistent with the current NARC objective treating same-class different-sample pairs as negatives.

2.4 Practical Caveats

Two implementation details may further limit NARC in the current setup:

- the view builder currently uses `tx` masking rather than richer multi-receiver views;
- the paper-aligned runs use `sampler = none`, not class-balanced contrastive batches.

These factors likely make the current instance-level NARC objective noisier than the class-aware SupCon objective.

3 Future Steps

1. Look more deeply into this week’s LOEO results, especially the class-wise behavior and the gap between the overall ranking and the best per-class performance.
2. Arrange the next round of experiments, including multi-seed validation, macro-F1-based checkpoint selection, and revised NARC settings.

4 Summary

This week established a clear result from the comparable LOEO ablations: **SupCon only is still the strongest completed objective**. SupCon + NARC + AdaIN slightly improves over SupCon + NARC, but it does not surpass **SupCon only**. The overall pattern remains the same: objectives centered on class-consistent clustering generalize better than the current instance-centered NARC formulation.